COMSATS University Islamabad
Department of Computer Science
Course Syllabus

Course Information

Course Code: **CSC101** | Course Title: **Introduction to ICT**
Credit Hours: **3(2,1)** | Lecture Hours/Week: **2**
Lab Hours/Week: **3** | Pre-Requisites: **None**

Catalogue Description:

This course covers the basics of Information and Communications Technologies. Topics include: Overview of ICT; Computing Models; Computer Systems & Components; Number Systems & Computer Codes; System & Application Software; Introduction to Databases & Information Systems; Computer Networks & Internet; Security; Future Trends in ICT; Problem Solving Concepts; Program Development Lifecycle; Introduction to Python.

Text and Reference Books

Textbooks:

1. Understanding Computers: Today and Tomorrow, Comprehensive, Deborah Morley, Charles S. Parker, Cengage Learning, 2017.
2. Python Basics: A Practical Introduction to Python 3, David Amos, Dan Bader, Joanna Jablonski, and Fletcher Heisler, Real Python, 2021

Reference Books:

1. Foundations of Computer Science, Forouzan, B., McGraw-Hill, 2017.
2. Starting Out with Python, Gaddis, T., Addison-Wesley, 2016.
3. Problem Solving & Programming, Sprankle, M., Hubbard, J., Prentice Hall, 2012.

Week wise Plan:

Lecture #	CDF Unit #	Topics Covered	Reading Material
1.	1	Basic Definition of Information Technology (IT), Computer Technology & Communications Technology, Role of IT in Society, Computing Models: Turing Model, Von Neumann Model; Input Devices: Keyboard, Pointing & Touch Devices, Game Controllers, Optical Input Devices, Audio Visual Devices; Output Devices: Monitors, Audio Output, and Printers & Plotters.	Morley: Ch1,Ch4
2.	1	How Computer Process Data; Central Processing Unit: Control Unit, Arithmetic & Logic Unit, System Clock & Machine Cycle; Memory: Volatile & Non-Volatile, Flash Memory, Registers, Cache Memory; Bus & Types, and Ports.	Morley: Ch2
3.	1	Storage Devices: Magnetic Storage Devices, Optical Storage Devices, Solid State Storage Devices, and Data Read & Write Process.	Morley: Ch3
4.	2	How Computer Represent Data: Number Systems, Bits & Bytes, and Text Codes.	Morley: Ch2
5.	2	Number Systems: Binary, Octal, Hexadecimal, and Conversions.	Morley: Ch2
6.	2	Number Systems Arithmetic: Addition & Subtraction of Decimal, Binary, Octal, and Hexadecimal Number System.	Ref. Material
7.	3	Computer Software: System Software, Application Software, Operating System (OS): Basic Purpose of OS, Types of OS,	Morley: Ch5, Ch6

		Operating System Functions: Providing User Interface, Running Programs, Managing Hardware, Utility Programs, PC Operating System, Network Operating System, and Embedded Operating System.	
8.	3	Database: What is a Database? Advantages & Disadvantages of the DBMS Approach, Data Concepts & Characteristics, Data Definition, Data Hierarchy Entities & Entity Relationships, Data Integrity, Security & Privacy, and Centralized vs. Distributed Database Systems.	Morley: Ch12
9.	3	Networking Concepts; Networking Applications: Telecom Industry, Videoconferencing; Network Characteristics: Types of networks, Network Topologies, Network Architectures, and Network Size & Coverage Area.	Morley: Ch7
10.	3	Data Transmission Characteristics: Bandwidth, Analog vs. Digital Signals, Transmission Type & Timing, Delivery Method; Networking Media: Wired Networking Media & Wireless Networking Media; Communications Protocols & Networking Standards; and Networking Hardware.	Morley: Ch7
11.	3	Basic Security Concepts: Threats to User: Identity Theft, Loss of Privacy, Online Spying Tools, Online Theft, Online Fraud, & Other Dot Cons; Threats to Hardware: Power Related Threats, Hardware Loss, Hardware Damage, System Failure, Unauthorized Access & Unauthorized Use, Natural Disasters; Threats to Data: Malware, Virus & Malicious Program, Cybercrime, Cyber-Terrorism, and Counter Measures.	Morley: Ch9
12.	3	Information Systems & System Development: Need for System Development Enterprise Architecture, Business Intelligence (BI), Users of Information Systems; Types of Information Systems; Responsibility for System Development; System Development Life Cycle (SDLC); and Approaches to System Development.	Morley: Ch10
13.	3	E-commerce: Overview, E-commerce Basics, Business Applications, Global Trends; Impacts: Impacts on Market & Retailers, Impact on Supply Chain Management, Impact on Employment, Impact on Customers, Impact on Environment, Impact on Traditional Retail; and Distribution Channels.	Ref. Material
14.	3	Cloud Computing: Overview, Advantages & Disadvantages, Characteristics; Service Models; Deployment Models; and Security & Privacy.	Ref. Material
15.	3	IoT: Applications; Trends, Characteristics & IoT Adoption Barriers; Augmented Reality (AR): Technology, Possible Applications, Remote Collaboration, and Danger of AR.	Ref. Material
16.	4	General Problem-Solving Concepts: Types of Problems; Problem Solving with **Sequential Structure**, Discussion & Practice of Pseudocodes, and Generating Flowcharts.	Ref. Material
17.	**Mid Term Exam**		
18.			
19.	4	Problem Solving with **Decision Structure**, Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Ref. Material
20.	4	Problem Solving with **Repetition Structure**, Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Ref. Material
21.	4	Problem Solving with **Procedure, Functions & Lists**, Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Ref. Material

22.	5	What is a Computer Program? Introduction to Python, Indentation, Control Structures, and First Python Program.	Amos: Ch3
23.	5	Variables: Integer Values, Variables & Assignment, Identifiers, Floating-Point Values, User Input, ***eval*** Function, Controlling ***print*** Function, and Difference between Expressions & Statements in Python.	Amos: Ch5
24.	5	Operators & Operands; Variable Comparison, Logical & Boolean Operations, Conditional Statements, and Order of Precedence.	Amos: Ch8
25.	5	Nested Conditionals, Multi-way Decision Statements, Conditional Expressions, Errors in Conditional Statements, and ***else-if*** statement vs ***if-if*** Statement.	Amos: Ch8, Ref. Material
26.	5	Strings, Control Codes within Strings, Slicing & Indexing Strings, Concatenation, and Arithmetic Operators with Strings.	Amos: Ch4
27.	5	***While*** Loop, Conditionals within a Loop, Counting using ***while*** Loop, and ***break & continue*** Statements.	Amos: Ch6, Ref. Material
28.	5	***For*** Loop, Conditionals within a Loop, Counting Using ***for*** Loop, and ***in*** Operator with ***for*** Loop.	Amos: Ch6, Ref. Material
29.	5	Nesting ***while*** Loop, Nesting ***for*** Loop, Nesting ***while*** inside ***for*** Loop, and Nesting ***for*** Inside ***while*** Loop.	Amos: Ch6, Ref. Material
30.	5	Functions: Predefined vs User Defined Function, Using Predefined Functions, Standard Mathematical Functions, and Random Numbers.	Amos: Ch6, Ref. Material
31.	5	User Defined Functions: Function Basics, Defining Functions, Calling Functions, Defining Main Function, and Parameter Passing.	Amos: Ch6, Ref. Material
32.	5	Void Function vs Value Returning Function, Fruitful Functions; Incremental Development, and Boolean Functions.	Amos: Ch6, Ref. Material
Final Term Exam			

Student Outcomes (SOs)

S.#	Description
1	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
2	Identify, formulate, research literature, and solve *complex* computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
4	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to *complex* computing activities, with an understanding of the limitations

Course Learning Outcomes (CLOs)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO's for Theory				
CLO-1	1	Explain the basic computing models and related hardware.	*Understanding*	1
CLO-2	2	Work out with different number systems and codes.	*Applying*	1,2
CLO-3	3	Describe the fundamental concepts of ICT domains.	*Understanding*	1
CLO-4	4	Solve computing problems using problem-solving process.	*Applying*	1,2,4
CLO-5	5	Describe the concepts of variables, conditional, repetitive structures, and functions using a	*Understanding*	1,2

		programing language.		
CLO's for Lab				
CLO-6	4	Demonstrate the usage of productivity software, problem solving tool, and web technology by performing appropriate tasks.	*Applying*	4
CLO-7	5	Implement an algorithm in a programming language to solve a simple problem.	*Applying*	2,4

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6	CLO-7
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-	-
Assignments	-	-	Assignment 1&2	Assignment 3	Assignment 4	Lab Assignment	Lab Assignment
Mid Term Exam	Mid Term Exam			-	-	-	-
Final Term Exam	Final Term Exam					-	
Project	-	-	-	-	-	-	Lab Project

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Total
Theory (T)	15	10	25	50	100
Lab (L)	-	25	25	50	100
Final Marks (T+L)	**(T/100) *67 + (L/100) *33**				

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	**A**	**A-**	**B+**	**B**	**B-**	**C+**	**C**	**C-**	**D+**	**D**	**F**
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.